

PAPER_1574_R_SUN_OVER_R_EARTH_109

Daniel T. Murphy

$$\text{PAPER_1574} \text{ — } R_{\blacksquare}/R_{\oplus} \text{ Ratio} = R_{\blacksquare}/R_{\oplus} = SO_5^2 + N_CH = 100 + 9 = 109$$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Astronomy

Date: June 18, 2026

Status: CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Z_S577 (Astronomy Unified Proof Set) lists $R_{\blacksquare}/R_{\oplus}$ Ratio as an EXACT-tier closure:

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$$R_{\blacksquare}/R_{\oplus} = SO_5^2 + N_CH = 100 + 9 = 109$$

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UQFF Closed Identity

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$$R_{\blacksquare}/R_{\oplus} = SO_5^2 + N_CH = 100 + 9 = 109 \text{ EXACT}$$

``

Pure integer-primitive product with zero fitted constants.

Physical Interpretation

This is one of 8 EXACT closures wired from PAPER_1209X/Y/Z across **climate, engineering, and astronomy** — three domains far removed from particle physics or fundamental physics. Yet each observable reduces to a simple integer-primitive expression.

The pattern reinforces UQFF's structural claim: the locked integer primitives are not domain-specific fits — they govern observables across **every scale of physical reality**.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating that the framework's predictive economy extends to engineering and engineering-adjacent observables (not just particle physics or cosmology).

Reference

- Source: PAPER_1209Z_S577
- Related: PAPER_1209X/Y/Z unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "r_sun_over_r_earth_109"})`

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